MIL-A-46104(MR)
26 October 1965

MILITARY SPECIFICATION

ALUMINUM ALLOY BAR, ROD, SHAPES, AND TUBE, EXTRUDED, 6070

1. SCOPE

1.1 Scope. This specification covers the specific requirements for aluminum alloy 6070 bar, rod, shapes, and tube produced by extrusion.

1.2 Classification.

1.2.1 Composition and tempers. Bar, rod, shapes, and tube covered by this specification shall be furnished in the composition shown in table I and in the following tempers, as specified (see 6.2).

T6 - Solution heat-treated and artificially aged.

T6510 - Solution heat treated, stress relieved by stretching to produce a permanent set of 1-1/2 percent nominal, but not less than 1 percent nor more than 3 percent, and artificially aged. Material shall receive no further straightening after stretching.

T6511 - Solution heat treated, stress relieved by stretching to produce a permanent set of 1-1/2 percent nominal, but not less than 1 percent nor more than 3 percent, and artificially aged. Material may receive minor straightening after stretching.

FSC 4710, 9530, 9540

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:52Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

STANDARDS

FEDERAL

Fed. Test Method Std. No. 151 - Metals; Test Methods
Fed. Std. No. 184 - Identification Marking of Aluminum, Magnesium and Titanium
Fed. Std. No. 245 - Tolerances for Aluminum Alloy and Magnesium Alloy Wrought Products

MILITARY

MIL-STD-649 - Aluminum and Magnesium Products: Preparation for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Chemical composition. The material shall conform to the chemical requirements shown in table I.

Table I - Chemical composition

	Percent	
	Minimum	Maximum
Silicon	1.0	1.7
Iron	-	0.50
Copper	0.15	0.40
Manganese	0.40	1.0
Magnesium	0.50	1.2
Chromium	-	0.10
Zinc	-	0.25
Titanium	-	0.15
Other elements, each	-	0.05
Other elements, total	-	0.15
Aluminum	Remainder	

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:52Z
Check the source to verify that this is the current version before use.

3.1.1 Analysis shall be made regularly only for the elements specifically mentioned in table I. If, however, the presence of the other elements is indicated in the course of routine analysis, further analysis shall be made to determine that the total of these other elements is not in excess of 0.50 percent.

3.2 Mechanical properties. Unless otherwise specified in the contract or order, the mechanical properties in the direction of extrusion shall conform to the requirements specified in table II. The elongation shall not be applicable to the following:

(a) Material of such dimensions that a standard test specimen cannot be taken in accordance with Fed. Test Method Std. No. 151, and of such shape that it cannot be satisfactorily tested in full section.

(b) Material thinner than 0.062 inch (nominal).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:52Z
Check the source to verify that this is the current version before use.

Table II - Mechanical properties

Temper	Thickness of thickest section	Tensile strength, psi, min.	Yield strength at 0.2 percent offset or at extension indicated		Elongation[1] (minimum) in 2 inches or 4 times diameter
			psi, Minimum	Extension under load	
T6, T6510, T6511	Up thru 2.999[2]	48,000	45,000	0.0065	6

[1]D represents specimen diameter.
[2]Maximum cross-sectional area - 32 sq. in.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:52Z
Check the source to verify that this is the current version before use.

3.3 Dimensional tolerances. Unless otherwise specified in the contract or order, tolerances shall be in accordance with Fed. Std. No. 245.

3.4 Identification marking. When specified in the contract or order, item identification marking shall be in accordance with Fed. Std. No. 184 (see 6.4).

3.5 Workmanship. The rod, bar, shapes and tube shall be uniform in quality and condition, clean, sound, smooth, and free from hard spots, pipes, laps, cracks, kinks, seams, damaged ends, and other injurious defects within the limits consistent with the best commercial practice. Discoloration caused by heat treatment will not be cause for rejection.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot. A lot shall consist of an identifiable quantity of bar, rod, shapes, or tube of the same size and temper submitted for inspection at one time.

4.3 Sampling.

4.3.1 For chemical analysis.

4.3.1.1 Ingot analysis. At least one sample shall be taken from each group of ingots of the same alloy poured simultaneously from the same source of molten metal by the producer and analyzed to determine conformance with 3.1. Ingots not conforming to the requirements of this specification shall be rejected. Complete ingot analysis records shall be made available.

4.3.1.2 Finished product analysis. Unless compliance with 4.3.1.1 is established, an analysis shall be made for each 4,000 pounds or less of material comprising the lot, except that not more than one analysis will be required per piece.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:52Z
Check the source to verify that this is the current version before use.

4.3.2 For mechanical properties.

4.3.2.1 Tension test. From material having a nominal weight of less than one pound per lineal foot, one tension test sample shall be selected from each lot weighing 500 pounds or less; from lots weighing more than 500 pounds, one additional sample shall be taken for each 1,000 pounds or fraction thereof in excess of the first 500 pounds. From material having a nominal weight of one pound or more per lineal foot, one tension test sample shall be taken from each lot consisting of 500 feet, or less; from lots consisting of more than 500 feet, one additional sample shall be taken for each 1,000 feet or fraction thereof in excess of the first 500 feet. Only one tension test sample shall be taken from any one piece when more than one piece is available.

4.4 Examination.

4.4.1 Visual and dimensional. Each bar, rod, shape or tube shall be carefully examined to determine conformance to this specification with respect to workmanship. Sufficient spot checks shall be made to insure conformance to the tolerances specified. On approval of the procuring activity, a system of statistical quality control may be used for dimensional and workmanship examination.

4.4.2 Preparation for shipment. Examination of the preservation, packaging, packing, and marking for shipment shall be made for conformance to the requirements of section 5.

4.5 Tests.

4.5.1 Test specimens.

4.5.1.1 Tension tests. Tension test specimens shall conform to the requirements of method 211 of Fed. Test Method Std. No. 151. When practicable, the material should be tested in full section. For rod, bar, or shapes which are not tested in full section, the standard R1, R2, R3 or F2 type specimen shall be used. For tube less than 0.500 inch in wall thickness, which is not tested in full section, the standard T1 type specimen shall be used. For tube having a wall thickness of 0.500 inch or more, a test specimen of type R1, R2, or R3 shall be used. For material which cannot be tested in full section and from which a standard specimen cannot be obtained, a strip cut from the material shall be tested.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:52Z
Check the source to verify that this is the current version before use.

4.5.1.2 Location of test specimens.

4.5.1.2.1 Rod, bar, and shapes. For sections which are wholly or predominately symmetrical, the tension test specimens shall be taken from the locations specified in table III. For sections of which the predominant part is unsymmetrical, the specimen shall be taken from a location that most nearly satisfies the intent of table III. Unless otherwise specified in the contract or order, for odd-shaped sections, only the predominant part shall be tested.

Table III - Location of axis of specimen

Thickness, inches	Location of axis of specimen with respect to thickness (T) and width (W) of section		
	Thickness	Width	
		Equal to or less than 1.50 inches	Greater than 1.50 inches
Less than 0.500	T/2	-	-
0.500 to 1.500, incl.	T/2	W/2	W/4
Greater than 1.500	T/4	-	W/4

4.5.1.2.2 Tube. Tension test specimens shall be taken from the tube in the longitudinal direction. From tube in wall thickness 0.5 inch to 1.5 inches, inclusive, tension test specimens shall be taken with the axis midway between the inside diameter surfaces; from tube having wall thickness greater than 1.5 inches the axis of the tension test specimen shall be three fourths of the distance from one surface to the other.

4.5.2 Test procedures.

4.5.2.1 Chemical analysis. The samples selected in accordance with 4.3.1 shall be analyzed by the wet chemical method in accordance with method 111 of Fed. Test Method Std. No. 151 or the spectrochemical method in accordance with method 112 of Fed. Test Method Std. No. 151 to determine conformance with 3.1. A single analysis of a composite sample may be made. In case of dispute, analysis by the wet method (method 111) shall be the basis for acceptance.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:52Z
Check the source to verify that this is the current version before use.

4.5.2.2 Tension tests. Specimens from samples selected in accordance with 4.3.2.1 shall be tested in accordance with method 211 of Fed. Test Method Std. No. 151.

4.5.2.2.1 Yield strength. Yield strength shall be determined by the offset method or the extension-under-load method in accordance with method 211 of Fed. Test Method Std. No. 151. In case of dispute, the offset method shall be the basis for acceptance.

4.6 Rejection.

4.6.1 Examination defects. Any sample unit having one or more defects shall be rejected.

4.6.2 Test failures. A lot shall be rejected for failure to meet any of the test requirements when tested in accordance with 4.5, subject to the provisions of the section on "Rejection and Retests" of Fed. Test Method Std. No. 151.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing. All bar, rod, shapes and tube shall be preserved and packaged in accordance with the requirements of MIL-STD-649. The procuring activity will specify the levels of packaging required. Unless otherwise specified in the contract or order, packaging shall be level C (see 6.2).

5.2 Marking. In addition to any special markings required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-649.

6. NOTES

6.1 Intended use. This alloy is intended for use where good strength and weldability are required along with good corrosion resistance and workability equivalent to other presently used structural alloys. The T6510 temper is intended primarily to provide extruded material with low residual stresses and consequent minimum distortion during machining. The T6511 temper is straightened T6510 material and may have unknown residual stresses and may or may not distort during machining.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:52Z
Check the source to verify that this is the current version before use.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Form and temper required (see 1.2.1).
(c) Dimensions required.
(d) Requirements for sizes not specifically covered.
(e) Levels of preservation, packaging, and packing required, if other than level C (see 5.1).

6.3 Heat treatment. Solution heat treatment of 6070 bar, rod, shapes, and tube is normally accomplished by quenching in water or air from the extrusion temperature at the extrusion die. The T6 temper may be obtained by artificial aging for 17 to 19 hours at a temperature of 320° ± 10°F or for 7 to 9 hours at a temperature of 350° ± 10°F. Severe forming operations can be accomplished more readily prior to artificial aging.

6.4 The requirements for item identification marking (see 3.4) and for packaging and packing and marking for shipment (see 5.1 and 5.2) specified herein apply to direct shipment for Government activities and apply also where specified, to contracts or orders bewteen the manufacturer and the Government prime contractor.

Custodian:
Army - MR

Preparing activity:
Army - MR

Review activities:
Army - MR, MU

Project No. 9530-A018

Review/User information is current as of the date of this document; draft circulation should be based on the information in the current DODISS

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:52Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil - Downloaded: 2026-10-22T04:52Z
Check the source to verify that this is the current version before use.